

Qatar-1b

R-2034

Benchmark run · workflow W8-benchmark-deep · record deep-bench_Qatar-1_121208 · created 2026-08-08T01:35:53+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2456269.603619 +/- 7.2e-05 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.175 +/- 0.013
Transit depth (Rp/Rs)^2	3.07 +/- 0.46 [%]
Orbital Inclination (inc)	83.47 +/- 0.53
Airmass coefficient 1 (a1)	1.056 +/- 0.012
Airmass coefficient 2 (a2)	-0.028 +/- 0.012
Scatter in the residuals of the lightcurve fit is	1.16 %
Best Comparison Star	#2 - [297, 422]
Optimal Aperture	3.02
Optimal Annulus	6.65
Transit Duration (day)	0.0651 +/- 0.0019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.175 $\pm$ 0.013 vs 0.1463 (deviation 2.0 σ)
Duration fit vs published	0.0651 d vs 0.0692 d (deviation 1.94 σ)
Mid-time O-C	-0.2 min
Depth SNR	12.2
Residual scatter	1.16 %

Light curve

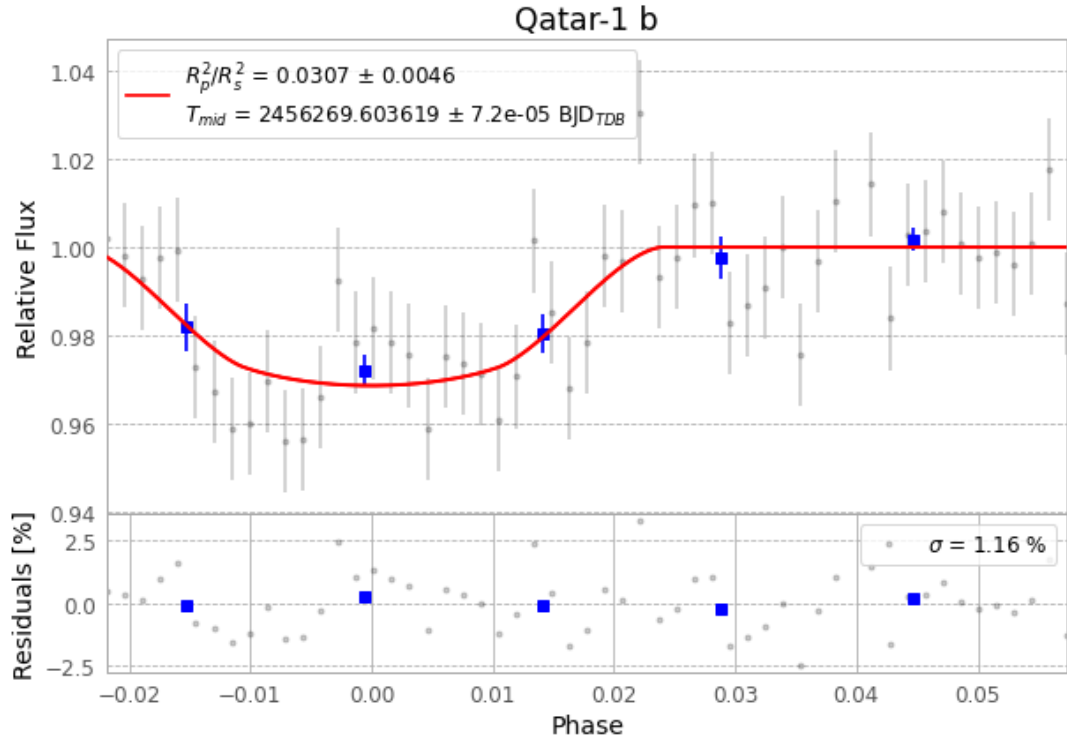


Figure 1 — FinalLightCurve_Qatar-1 b_2012-12-07.png (SHA-256 0dfb65c9389179d5...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [335, 252], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 5a276b9dff7fafd3...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit fd74a91

Data provenance

55 FITS frames (36 MB), Qatar-1121208014212.FITS → Qatar-1121208042412.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-Qatar-1-121208.log (3f5a9b93767d...), FinalLightCurve_Qatar-1 b_2012-12-07.png (0dfb65c93891...), FinalLightCurve_Qatar-1 b_2012-12-07.csv (60e757496604...)

Compute resources

Wall time 128.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-08 01:35Z.